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A systematic review of health-related quality of life outcomes in psychosocial intervention trials for adolescent and young adult cancer survivors

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Abstract

Background: The National Cancer Institute has catalyzed research in adolescent and young adult (AYA) oncology by identifying the need for supportive care intervention studies and psychometrically robust health-related quality of life (HRQOL) measures. We evaluated progress toward these goals (1) examining changes in the number of registered psychosocial intervention trials being conducted with AYAs over time; (2) determining what domains of HRQOL were assessed across these intervention trials; and (3) identifying the most frequently used measures of HRQOL.

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- **Karly M. Murphy, PhD:** Conceptualization, Methodology, Validation, Formal Analysis, Investigation, Data curation, Writing – Original draft preparation, Visualization, Project administration
- **Elizabeth Siembida, PhD, MPH:** Conceptualization, Validation, Investigation, Writing – Review & editing
- **Nancy Lau, PhD:** Conceptualization, Validation, Investigation, Writing – Review & editing
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Methods: We conducted a systematic review of psychosocial intervention trials for AYAs registered on ClinicalTrials.gov from 2007–2021. Following identification of relevant trials, we extracted the outcome measures and determined whether they were measures of HRQOL and which HRQOL domains were evaluated. Descriptive statistics were used to summarize trial and outcome characteristics.

Results: We identified 93 studies that met our inclusion criteria and 326 HRQOL outcomes across studies. The average number of clinical trials conducted annually has increased from 2 (SD = 1) during the years of 2007–2014 to 11 (SD = 4) during the years of 2015–2021. 19 trials (20.4%) did not include a measure of HRQOL. HRQOL measures varied widely, and most evaluated psychological and physical domains. Of the 9 measures used 5+ times, none were developed to cover the full AYA age spectrum.

Conclusions: This review demonstrated that the number of AYA psychosocial intervention trials conducted annually has increased. However, it also revealed several important areas for additional work including: (1) ensuring psychosocial trials include HRQOL measures; (2) increasing the frequency of evaluation of underrepresented domains of HRQOL (e.g., body image, fertility/sexuality and spiritual); and (3) improving the validity and standardization of measures used to evaluate domains of HRQOL across AYA-focused trials to improve the field's ability to compare the impact of different psychosocial interventions on HRQOL outcomes.

Keywords

Psychosocial Intervention; Adolescent; Young Adult; Psycho-Oncology; Quality of Life; Systematic Review; Patient Reported Outcome Measures

Introduction/Background

In 2006, the National Cancer Institute in conjunction with Livestrong™ released a report from the Adolescent and Young Adult (AYA) Oncology Progress Review Group. This landmark report defined AYAs as individuals diagnosed with cancer between the ages of 15 and 39, identified them as an understudied and medically underserved population, and called for additional research to improve care delivery for AYAs.[1] Since the release of this report, there has been a proliferation of research focused on AYAs and the five-year relative survival rate of AYAs for all cancers combined has increased from 71% in the mid-1970s to 83–86% in 2020.[2,3] Consistent with recommendations from this report and improving survival rates, a subset of this research has focused on the health-related quality of life (HRQOL) of AYAs. Interventions that target HRQOL in this population are critical to ensure that AYAs are able to live well following cancer diagnosis and treatment.[4,5]

Though there are multiple, overlapping definitions of HRQOL,[6] here we define it as the extent to which one's usual or expected physical, mental, and social well-being are affected by a medical condition or its treatment.[7] This concept is particularly important for AYAs given that a diagnosis of cancer is non-normative for individuals in this age-group, and often disrupts achievement of key developmental milestones. In recent years it has been well established that these disruptions result in lower levels of HRQOL among AYAs with cancer when compared to either their healthy peers or older cancer survivors.[8–11] Furthermore,

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there is an elevated prevalence of chronic health conditions among AYA cancer survivors, which can further reduce HRQOL.[12]

Despite evidence for deficits in HRQOL among AYAs, there are few measures of HRQOL that have been developed for and/or validated in this population. As a result, measures developed for other populations are often used to evaluate HRQOL in AYAs. This is potentially problematic. Existing measures of HRQOL developed for other populations may not be responsive to change in the AYA population. This may result in a lack of precision in identifying symptomatic or functional changes as well as in our ability to assess the efficacy of potential interventions.[13] Additionally, there are domains of HRQOL with unique relevance to this population that are often not evaluated by broadly used measures of HRQOL.[13] For example, given that AYAs are of child-bearing age and often treated with therapeutics that can impact fertility, it is important to assess fertility-related concerns as a component of HRQOL. This is reflected in models of HRQOL for young adult cancer survivors, such as the expanded City of Hope Framework recommended by Nightingale and colleagues.[14] The City of Hope Framework consists of the core HRQOL domains of physical, psychological, social, and spiritual well-being. The expanded framework incorporates domains uniquely relevant for young cancer survivors such as body appearance, resilience, and fertility/sexual functioning. We chose to use this expanded model of HRQOL in young cancer survivors to ensure our evaluation captured the breadth of HRQOL domains relevant to AYAs.

In 2016, an update to the 2006 report outlined several next steps for evaluating and improving HRQOL in AYA oncology.[4] These recommendations included conducting supportive care intervention studies to address the "physical, psychological, and social health deficits among AYAs with cancer" as well as the development of "valid, reliable, developmentally relevant, and psychometrically robust measures of HRQOL…that cross the age spectrum and allow for studies of the full AYA age range." The goal of the present study is to evaluate progress toward these recommendations by examining the measures of HRQOL used in psychosocial intervention trials for AYAs registered on ClinicalTrials.gov. Specifically, we sought to (1) evaluate changes in the number of registered psychosocial intervention trials being conducted with AYAs over time; (2) determine what domains of HRQOL were assessed across these intervention trials; and (3) identify the most frequently used measures of HRQOL.

Methods

Trial Extraction.

We conducted a search of ClinicalTrials.gov from January 1st 2007 to November 1st, 2021. We searched for intervention studies that included children and/or adults using the following search terms: adolescent OR adolescence OR teen OR boys OR girls OR teenagers OR youth OR youths OR "young adult" OR "young women" OR "young men" OR "young patients" OR AYA. Additionally, the Advanced Search feature of ClinicalTrials.gov was used to limit condition or disease to: cancer OR neoplasm OR tumor OR malignancy. Similarly, intervention/treatment was limited to: Behavioral OR Device OR Procedure OR Other. Studies with a recruitment status of "Terminated" or "Withdrawn" were excluded, as

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were studies with a start date prior to January 1st, 2007. The year 2007 was used as it was the year after the NCI issued its report "Closing the Gap: Research and Care Imperatives for Adolescents and Young Adults with Cancer" which called for the development of new AYA-specific psychosocial interventions as well as the evaluation of existing interventions.[1]

Initial Review of Trials.

An initial review of the extracted trials was conducted by KM to exclude studies based on age-related inclusion/exclusion criteria as reported in the ClinicalTrials.gov record. As these exclusions were made based on objective information, no coding requiring multiple researchers was conducted. Specifically, studies were excluded if (1) the trial excluded participants between the ages of 15 and 39 years, or (2) the trial included participants younger than 10 years old or older than 45 years old. While AYAs may represent a small proportion of participants in trials with broader age ranges, we elected to utilize these age cut-offs to ensure the trial's target population consisted primarily of AYAs since the goal of our study was to understand HRQOL assessment in this population. This expanded age-range of 10–45 years was chosen because there is significant variation in how researchers define adolescents and young adults.[15] We chose age 10 as our lower limit because adolescence has historically been defined as ages 10–19.[16] Our choice of 45 as the upper limit was based on the importance of fertility-focused interventions for this age group. Many such interventions include participants up to age 45 as this is widely accepted to be the upper limit of reproductive age for females.[17]

Trial Screening.

Following initial review, trials were screened by two coders (KM and ES, NL, or JS) for inclusion. Trials were considered for inclusion if they met the following criteria: (1) participants were cancer patients/survivors; (2) participants were diagnosed with cancer between the ages of 10 and 45; and (3) the intervention being tested met our definition of psychosocial (any non-pharmacologic, therapeutic approaches that target thoughts, feelings, or behaviors). Inter-rater reliability was calculated as the percentage of trials which the two coders agreed on whether to include or exclude it. Discrepancies were resolved via discussion with all coders (KM, ES, NL, and JS) including review of the trial record on ClinicalTrials.gov until consensus was reached. For trials included in the final sample, a database of trial information was created. This database included year of initiation; study status; availability of results on ClinicalTrials.gov; randomization; planned/actual enrollment; cancer diagnoses included; country; age-range of eligible participants; and sex of eligible participants. From these data and additional review of ClinicalTrials.gov records, we were able to determine AYA age sub-groups included (adolescents ages 10–17, emerging adults ages 18–25, and/or young adults ages 26–45)[18] and intervention type (individual, dyadic, family, or group). Additionally, we categorized trials by study design as single-arm pilots that evaluated feasibility/acceptability and/or proof of concept, pilot randomized controlled trials designed to evaluate feasibility/acceptability, and randomized controlled trials designed to evaluate efficacy. Studies which did not fit within any of these categories (e.g. cross-over assignment) were categorized as "other," (see Supplemental Materials for categorization by study).

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Outcome Extraction and Screening.

Once the trials to be included in this study had been determined, outcome measures were manually extracted by KM from the relevant section of the ClinicalTrials.gov record for each trial. Data extracted included the level of outcome (primary, secondary, or other), the name of the measure (if provided), and the investigator's description of the measure. Outcome measures were coded by two investigators (KM and ES, NL, or JS) to determine whether they were a measure of one or more of the domains of HRQOL in the expanded City of Hope framework.[14] If so, the investigator also coded which domain(s) the measure evaluated (physical, social, psychological, spiritual, body appearance, resilience, and/or fertility/sexual functioning). An "Other" option was also available if the coder determined that the HRQOL measure (or a subscale of the measure) did not fit in one of these domains. Discrepancies were resolved via discussion with all coders (KM, ES, NL, and JS) including review of the trial record on ClinicalTrials.gov and the measure (if identifiable and available) until consensus was reached. If a measure was not clearly identified in the ClinicalTrials.gov record and the description available in the record did not provide enough information for categorization, the measure was coded as "Not Enough Information" (NEI) and excluded from subsequent analyses.

Data Analysis.

Descriptive statistics were used to examine study and outcome characteristics. Study characteristics examined included the year of initiation and study design as reported on ClinicalTrials.gov. We also conducted a t-test to evaluate differences in the average number of AYA-focused psychosocial trials conducted from 2007–2014 v. 2015–2021. Study inclusion and exclusion criteria for age-group, disease type, and treatment status were summarized as well. We then calculated the proportion of trials that included HRQOL or one of its domains as a primary outcome and as any type of outcome (e.g., primary, secondary, or tertiary). For each domain of HRQOL, we calculated the number of times it was evaluated across all trials. Additionally, we identified and characterized the most used measures for each domain.

Results

Trial Identification.

Figure 1 shows the flow of trial identification and screening. Initially, 4714 studies were identified. Prior to screening, 4217 of these were excluded because they did not include participants between the ages of 15 and 39 (n = 337) or included participants well outside the AYA age-range (less than 10 years old or older than 45 n = 3880). The remaining 497 studies were screened for inclusion/exclusion by two coders. Inter-rater reliability was 90%, and discrepancies were resolved by discussion. Ultimately, 403 additional studies were determined to be ineligible for inclusion because they did not focus on cancer patients or survivors (n = 336), allowed patients to have been diagnosed with cancer during childhood (n = 20), or the intervention delivered did not meet our definition of psychosocial (n = 48). The final sample consisted of 93 trials.

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Characteristics of Included Trials.

Of the 93 studies we identified, 26 (28.0%) were single-arm pilots, 28 (30.1%) were pilot RCTs designed to evaluate feasibility/acceptability, 31 (33.3%) were RCTs designed to evaluate efficacy, and 8 (8.6%) did not fit into any of these categories (e.g., cross-over assignment, full factorial trial). Most studies were reported to be currently active (n= 46). Though 39 (41.9%) of the trials had been completed, only 5 (12.8% of completed trials) had results available on ClinicalTrials.gov, (i.e., the researchers had added their findings to the ClinicalTrials.gov database) and only 14 (35.9%) had publications identified by ClinicalTrials.gov Identifier (NCT Number) in Medline. On average, completed single-arm pilots enrolled 22 participants (SD = 10.2; range = 5 to 43); completed pilot RCTs enrolled 53 participants (SD = 39.7, range = 10 to 151) and completed RCTs enrolled 169 participants (SD = 152.7; range = 15 to 467).

With regard to inclusion/exclusion criteria, the majority of trials were agnostic in the matter of disease site (n = 71, 76.3%), though some trials were focused on AYAs with specific cancer sites (e.g., breast, n = 13, 14.0%). Similarly, the majority of trials included participants of all sexes (n = 73, 78.5%); however, some interventions were sex-specific as a result of diagnostic inclusion criteria (e.g. a study of testicular cancer only included males) or intervention content (e.g., a fertility preservation decision making intervention only focused on the concerns of females). Just over half of the studies we identified (n = 51; 54.8%) only included participants between the ages of 15 and 39, the range defined by the Adolescent and Young Adult Oncology Progress Review Group.[1] Minimum age of inclusion ranged from 10 to 32, while maximum age of inclusion ranged from 17 to 45. Additional details regarding the characteristics of included trials are presented in Table 1.

Characteristics of HRQOL Outcomes.—We manually extracted 982 outcomes from the 93 included trials. Using the procedures described above, 326 of these outcomes were determined to be measures of HRQOL. There were 12 measures of HRQOL for which there was not enough information to determine domain coverage, and these measures were excluded from all subsequent analyses, resulting in a final count of 314 HRQOL outcomes.

Aim 1: Evaluate changes in the number of registered psychosocial intervention trials being conducted with AYAs over time.

The average number of clinical trials conducted annually has increased from 2 (SD = 1) during the years of 2007–2014 to 11 (SD = 4) during the years of 2015–2021. The number of AYA-focused psychosocial intervention trials by year and study design are presented in Figure 2. Of the identified trials, 19 (20.4%) did not include a measure of HRQOL. A measure of HRQOL was the primary outcome in only 34 trials (36.6%). Since we would expect pilot trials to have a primary outcome focused on feasibility or acceptability, we also looked at the proportion of RCTs with a primary outcome that evaluated HRQOL; only 17 (54.8%) of RCTs included HRQOL or one of its domains as a primary outcome. However, 26 (83.9%) of the RCTs we identified did report measuring HRQOL across outcome levels (primary, secondary, or other).

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Aim 2: Determine domains of HRQOL assessed.

Across all outcome levels, the HRQOL domain most frequently measured was psychological (n = 159), followed closely by physical (n = 138). The social and resilience domains were also commonly measured (n = 89 and n = 52, respectively); however the spiritual (n = 14), body image (n = 15), and fertility/sexuality domains (n = 13) of HRQOL were rarely evaluated. This pattern broadly held true across primary, secondary, and other outcomes (see Figure 3).

Aim 3: Identify the measures most commonly used to assess HRQOL.

Overall, there was a high level of heterogeneity in the measures used across studies; we were able to identify 177 unique measures, 110 of which were used only in one study. At all outcome levels, the measures used 5+ times and their domain coverage are presented in Table 2. Notably, no measure was used more than nine times across all studies and outcome levels. Additionally, seven of these measures were developed for the adult population,[19–26] and only one was specifically developed for a cancer population.[27]

Discussion

This study is the first review of psychosocial intervention trials in AYA cancer patients/survivors registered on ClinicalTrials.gov. Our findings report that in the time since the publication of *Closing the gap: Research and care imperatives for adolescents and young adults with cancer* there has been an increase in the number of trials of psychosocial interventions for this population. This finding is consistent with other recent reviews that have evaluated the literature on AYA cancer survivors.[28,29] We also determined that 20.4% of psychosocial trials did not evaluate at least one domain of HRQOL. This proportion is lower than the proportion of AYA-relevant therapeutic trials that did not include a patient-reported outcome of HRQOL (79.3%) and comparable to the proportion of pediatric oncology behavioral supportive care trials that did not include an HRQOL measure (26.1%).[29,30] Finally, the most commonly used measures of HRQOL provide valid, reliable evaluations of physical, psychological, and social HRQOL in other populations, suggesting that researchers are attempting to use the best measures available. Overall, our findings indicate that meaningful progress has been made toward some of the recommended next steps in AYA oncology outlined in 2016.[4] However, this work has also allowed us to identify opportunities to improve future AYA psychosocial intervention trials in three major areas: (1) ensuring psychosocial trials include HRQOL measures; (2) increasing the frequency of evaluation of underrepresented domains of HRQOL (e.g., body image, fertility/sexuality and spiritual); and (3) improving the validity and standardization of measures used to evaluate domains of HRQOL across AYA-focused trials.

First, we determined that one in five psychosocial trials did not report a measure of any domain of HRQOL. Depending on the intervention target, it may not be appropriate to evaluate HRQOL or one of its domains as the primary outcome. That said, most psychosocial interventions would be expected to have some impact on HRQOL. For example, an RCT of an intervention that seeks to increase time spent engaging in moderate to vigorous physical activity (a primary outcome that is not a measure of HRQOL) may

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also be expected to result in improvements in HRQOL domains (e.g., fatigue in the physical domain, depressive symptoms in the psychological domain; see Caru et al.[31] for a recent review of physical activity studies in AYAs). Thus, not evaluating HRQOL as secondary or other outcomes in any number of psychosocial trials represents a missed opportunity in this understudied population.

Second, our review suggests that certain domains of HRQOL (i.e., psychological, physical) have been examined far more often than others (i.e., spiritual, body image, fertility/sexuality). This finding complements those recently reported by Engstrom et al. and Zhang et al. regarding a similar pattern in published literature on AYAs.[32,33] While psychological and physical domains of HRQOL are consistently reported as areas of need, so are social, spiritual, fertility/sexuality, body image, and other issues such as financial and professional concerns.[13,14,17,28,34–36] Taken together, these findings highlight the need for the use of measures that evaluate aspects of HRQOL that are particularly relevant to AYAs, into future AYA-focused research. We also found that few of the measures commonly used to evaluate HRQOL (2 of 9) were not specific to the needs and concerns of cancer survivors and (b) their reliability and validity has yet to be established across the full AYA age spectrum as recommended in the AYA oncology progress update.[4] Most available measures of HRQOL and its domains have been developed for either pediatric/adolescent populations or adult populations.[13] It is important to have common measures that can be used across the AYA age-range or to incorporate measures with established psychometric strategies for linking of scores between pediatric and adult versions such as the PROMIS physical health and emotional distress measures.[37,38] This will allow researchers to identify similarities and differences in symptoms and functioning across developmental sub-groups of AYAs.[18]

Finally, consistent with reviews of therapeutic trials,[30] supportive care trials in pediatric oncology,[29] and published studies focused on AYAs[32] our analyses also indicated that the measures used to assess HRQOL have varied widely across trials. In this study, the measures we identified as most used across trials have been shown to be psychometrically robust in adult non-cancer populations. Overall, the lack of consistency in the measures administered across trials without a standardized metric to inform crosswalks (i.e., linked scores) limits the field's ability to compare the impact of different psychosocial interventions on HRQOL outcomes.[39] At the same time, we recognize that interventions may target different outcomes depending on their focus, so a one-size-fits-all approach is not appropriate. One potential solution to this problem that has previously been suggested is the development of an AYA-specific core patient-centered outcome set.[40] If investigators were to use a core outcome set in future studies, the field will be able to compare and contrast the impact of psychosocial interventions on HRQOL. At the same time, having a core outcome set rather than a fixed battery ensures that investigators have the flexibility to choose HRQOL measures that are relevant to their research question and developmentally appropriate for research participants. In response to this call, the NCTN AYA Patient-Reported Outcomes Task Force recently recommended a core set of HRQOL domains and corresponding, developmentally relevant questionnaires for investigators to choose from depending on the goals of their studies.[41] These recommendations will assist researchers in identifying appropriate measures for their study and may ultimately result in improved standardization of measurement across intervention trials.

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Limitations.

This study makes several important contributions to the literature, but it has limitations. Most notably, the analyses we conducted were based only on the data available via ClinicalTrials.gov. As such, studies not registered in this US-based database were not captured in our evaluation. Similarly, information provided regarding the registered trials that have not yet been completed represents trial planning rather than the actual conduct of the trial. Furthermore, our review was limited to the information available as of November 1st, 2021 and may have been updated since that time. For example, we report that findings from NCT04150848 and NCT02832154 had not been published as of this date; however, publications are now available.[42,43] Additionally, our review focused on interventions targeting post-treatment survivors still within the AYA age range; as such studies focused on evaluating interventions for long-term survivors of AYA cancers (e.g. recent work by Janssen et al.)[44] were not included. Furthermore, the information we had about the interventions and outcomes were limited to what was provided by the investigator on this registry. At times, this limited our ability to identify additional study details relevant to our investigation. For example, domain coverage for some measures of HRQOL could not be determined because not enough information about the outcome was provided on the registry. Based on our experience using the ClinicalTrials.gov database for our study, we encourage investigators to provide more detailed, clear information about their study and the measures used to evaluate relevant outcomes when registering trials for the purposes of open science and accountability.

Conclusions.

This systematic review of AYA psychosocial intervention trials and their outcomes as registered on ClinicalTrials.gov demonstrated that the number of such trials has significantly increased over the past 15 years. However, the fact one in five of such trials did not include a measure of HRQOL represents a missed opportunity in this understudied and medically underserved population. Furthermore, this study revealed that more research is needed on domains of HRQOL that are particularly important to AYAs but rarely assessed such as body image and fertility/sexuality; the lack of measures of that evaluate these domains may be a key barrier to their inclusion in trials. As such, the development of measures that examine them will be an important next step. Moreover, the measures being used to evaluate HRQOL rarely cover the full age-range of AYAs and vary widely across studies. The creation of crosswalks between pediatric and adult measures[37,38] and different measures of the same construct[39] may help address these issues, as will the creation of an AYA-specific core outcomes set by the NCTN.[41]

Finally, our experience conducting this study revealed the importance of providing detailed information about outcomes when registering intervention studies on ClinicalTrials.gov as well as providing updated information about study status and findings. Effectively using tools such as ClinicalTrials.gov that promote greater transparency and scientific integrity will ultimately allow our field to close the research to practice gap and ensure widespread access to AYA-focused psychosocial interventions.

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Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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John M. Salsman, PhD is a health psychologist whose research focuses on maximizing the health-related quality of life of AYAs both during and after treatment.

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HIGHLIGHTS:

- The mean number of psychosocial trials for AYAs initiated annually increased to 11.
- 20% of trials did not include an HRQOL measure, representing a missed opportunity.
- HRQOL measures varied widely, limiting comparison across studies.
- Spirituality, body image, and fertility/sexuality domains were rarely measured.
- Of the 9 measures used 5+ times, 0 were developed to cover the AYA age spectrum.

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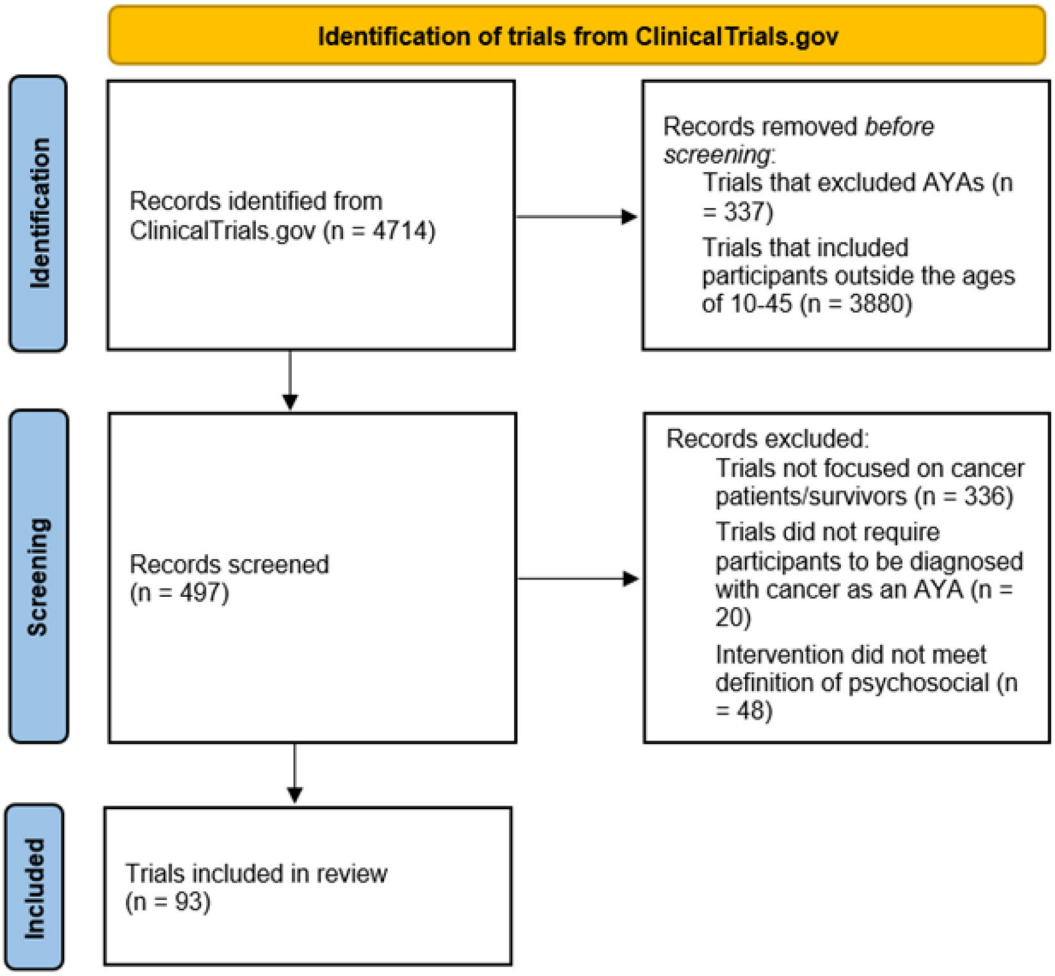

Figure 1.
Flowchart of trial identification and screening.

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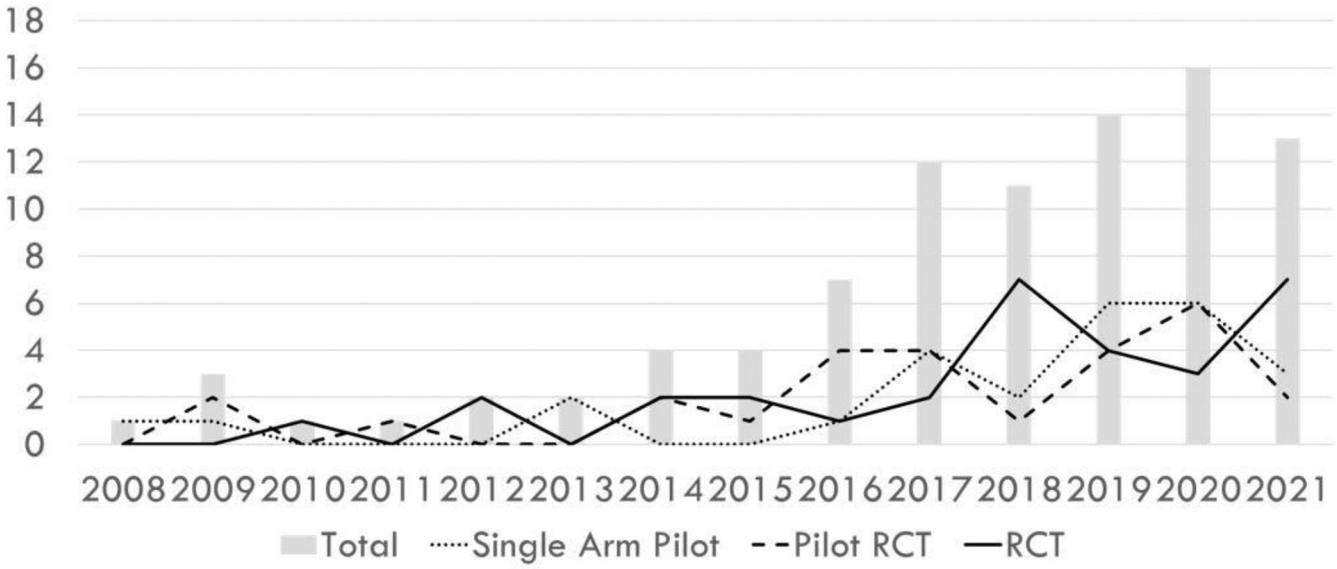

Figure 2.
Number of trials over time by study design and in total.

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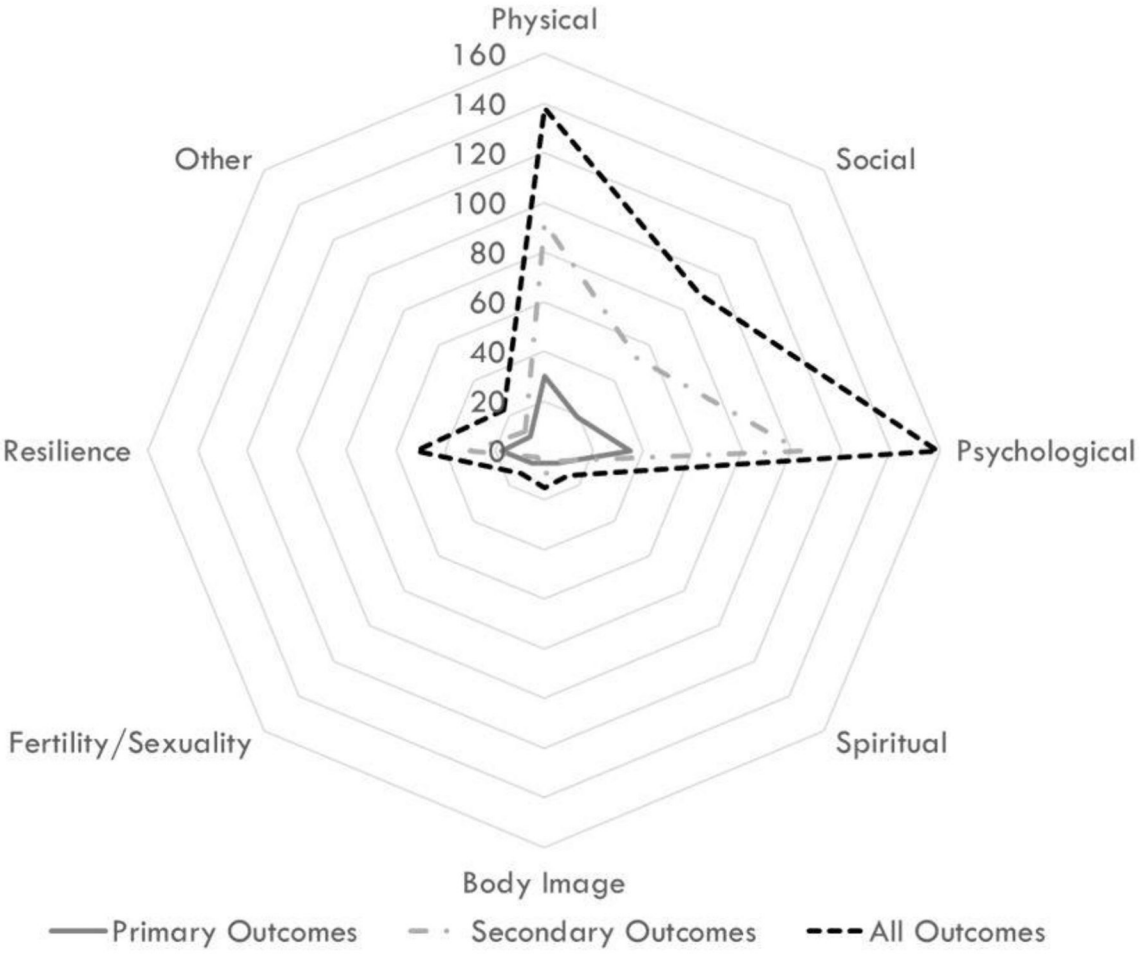

Figure 3.
Domain coverage by outcome level.

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Table 1.

Descriptive summary of study characteristics.

Study Design	**N**	**%**
Single-arm pilot	26	28.0
Pilot randomized trial	28	30.1
Randomized controlled trial	31	33.3
Other	8	8.6
Study Status	**N**	**%**
Active, not recruiting	6	6.5
Enrolling by invitation	1	1.1
Completed	39	41.9
Not yet recruiting	8	8.6
Recruiting	31	33.3
Suspended	2	2.2
Unknown	6	6.5
Age Inclusion Criteria	**M**	**SD**
Minimum	16.3	3.2
Maximum	34.2	9.0
Age-Groups Included*	**N**	**%**
Adolescents (ages 10–17)	44	47.3
Emerging Adults (ages 18–25)	80	86.0
Young Adults (ages 26–45)	67	72.0
Sexes Included	**N**	**%**
All	73	78.5
Female	17	18.3
Male	3	3.2
Cancer Diagnoses Included	**N**	**%**
Breast	13	14.0
Hematologic	4	4.3
Hematologic & solid tumor	71	76.3
Lower extremity	1	1.1
Oral cavity	1	1.1
Sarcoma	2	2.2
Testicular	1	1.1
Intervention Type	**N**	**%**
Individual	74	79.6
Dyadic	1	1.1
Family	2	2.2
Group	15	16.1
Unknown	1	1.1

*Percentages add up to more than 100% as studies often included multiple age-groups

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Table 2.

Measures of HRQOL used 5+ times in AYA focused psychosocial trials identified on ClinicalTrials.gov (2007–2021).

	P	S	O	Physical	Social	Psychological	Spiritual	Body Image	Fertility/ Sexuality	Resilience	Other
Hospital Anxiety and Depressio n Scale (HADS)[19]	3	5	1			Anxiety, Depression					
PROMIS® Anxiety[20]	2	6	0			Anxiety					
Functional Assessment of Cancer Therapy – General (FACT-G)[21]	1	6	1	Physical well-being, Functional well-being	Social/family well-being	Emotional well-being					
Pediatric Quality of Life Inventory (PedsQL 4.0)[45]	4	4	0	Physical	Social	Emotional					School
PROMIS® Fatigue[22]	1	6	0	Fatigue							
PROMIS® Depression[20,23]	2	5	0			Depression					
Pediatric Cancer Quality of Life Inventory (PedsQL 3.0 Cancer)[27]	2	4	0	Pain, Nausea	Communication	Procedural anxiety, Treatment anxiety, Worry, Cognitive problems		Perceived physical appearance			
RAND Medical Outcomes Study 36-Item Short Form Survey (SF-36)[24]	0	5	1	Physical functioning, Role limitations due to physical health, Energy/ fatigue, Pain, General health	Social functioning	Role limitations due to emotional problems, Emotional well-being					
Pittsburgh Sleep Quality Index (PSQI)[25]	1	3	1	Sleep quality							

Note: P indicates primary outcome, S indicates secondary outcome, and O indicates other outcome (as classified by study researchers on ClinicalTrials.gov)